Ioanna Gemou

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Education

Ph.D. in Computer Science — Aug. 2025–now
Brown University
Advisor: Ritambhara Singh

M.Sc. in Biomedical Engineering — Sep. 2023–July 2025
Technical University of Denmark
Advisors: Prof. Dimitrios Papadopoulos, Prof. Ritambhara Singh (Brown University)

B.Sc., M.Eng. in Electrical and Computer Engineering — Oct. 2018–July 2023
University of Patras
Advisors: Prof. Konstantinos Moustakas, Prof. Athanasios Lourbopoulos

Research Positions

Visiting Research Fellow — Aug. 2024–Feb. 2025
Brown University — Providence, RI, USA
Advisors: Nikos Vasilakis, Ritambhara Singh
Contributed to K-Paths (C1), a model-agnostic framework that extracts diverse, biologically meaningful reasoning paths from biomedical knowledge graphs to improve drug-drug interaction prediction and interpretability; led the end-to-end development of the Digital Pyramids Text data-processing pipeline, from system design through front-end integration to server deployment.

Research Assistant — Dec. 2022–May 2024
University of Athens — Athens, Greece
Advisor: Athanasios Lourbopoulos
Led the development and implementation of machine learning methods for neuroimaging and data analysis in neurological disorders, including deep-learning analysis of stroke on brain sections (A3).

Industry Positions

Electronics & Software Engineer — Sep. 2023–July 2024
Radiometer Medical A/S — Copenhagen, Denmark
Performed hardware and software verification testing on medical devices to ensure compliance with safety and performance standards; developed and executed automated test scripts, reducing manual testing time.

Software Engineer (Erasmus+ Placement) — June 2022–Aug. 2022
Bionos Biotech S.L. — Valencia, Spain
Contributed to computational pipelines for metabolomics and gene-expression analysis, including a Python/Selenium/Bash tool that retrieves metabolite data and groups animal and plant species by their metabolite profiles, and R and Python scripts that automate downstream gene-expression analysis.

Honors & Awards

Gerondelis Foundation Fellowship — 2026

Student/Travel Grants: KDD '25, NeurIPS '25 — 2025

1st Prize, BCI for Stroke Rehabilitation Competition, University of Patras — Nov. 2022

Publications

Conference Papers

C1. K-Paths: Reasoning over Graph Paths for Drug Repurposing and Drug Interaction Prediction. Tassallah Abdullahi, **Ioanna Gemou**, Nihal V. Nayak, Ghulam Murtaza, Stephen H. Bach, Carsten Eickhoff, Ritambhara Singh. **KDD '25**. [pdf] [code]

Workshop Papers

W2. LLM-Powered Graph Reasoning for Knowledge Discovery. **Ioanna Gemou**, Tassallah Abdullahi, Ritambhara Singh. **WiML @ NeurIPS '25**.

Conference Abstracts

A3. Deep-Learning-Network Analysis of Stroke on Brain Sections: The StrokeAnalyst-2. **Ioanna Gemou**, Panagiotis Kiousis, Angelos Pavlopoulos, Anastasia Papavasileiou, Uta Mamrak, Stefan Roth, Nikolaus Plesnila, Constantinos Pantos, Iordanis Mourouzis, Konstantinos Moustakas, Athanasios Lourbopoulos. **ESOC '26**. *European Stroke Journal* 11 (Suppl. 1).

Preprints

P1. A Geometric View of Counterfactual Behavior: Interaction of Boundary Proximity and Local Support. **Ioanna Gemou**, Matteo Gamba, Randall Balestriero, Ritambhara Singh. **arXiv 2606.04209**.

P2. A Proof-of-Concept for Explainable Disease Diagnosis Using Large Language Models and Answer Set Programming. **Ioanna Gemou**, Evangelos Lamprou. **arXiv 2512.23932**.

Talks

Guest Lectures

1. Multimodal Learning
   CS 582: Machine Learning for Bioinformatics, UIUC — Sep. 2026
   CSCI 1851: Machine Learning for Biology and Health, Brown University [slides] — Feb. 2026

Teaching

Machine Learning for Biology and Health (CSCI 1851) — Spring 2026
Brown University — Providence, RI, USA
Undergraduate level; enrollment: ~35; instructor: Prof. Ritambhara Singh
Created a series of six assignments guiding students from classical machine learning to deep learning on biomedical data, spanning heart-disease classification, epigenetic age prediction, RNA sequence classification with CNNs, autoencoders for single-cell RNA-seq, and graph neural networks for drug–disease and drug–drug interaction prediction [material].

Programming Instructor — Nov. 2021
Patras Codecamp — Patras, Greece

Service

Web & planning chair — 2025
Brown Systems Week — Providence, RI, USA

Web & publication chair — 2025
18th European Workshop on Systems Security (EuroSec 2025)

Languages

English (fluent), Greek (native), French (beginner), Spanish (beginner)

Last updated: November 12, 2026